LAB MODULE 2 CSE3023 – Web-based Application Development

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2.9 Exercise

In this lab, we will explore how to use DAO.

Source Code add_product.html:

```html
<html>
    <head>
        <title>Form</title>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
    </head>
    <body>
        <h2>Form</h2>
        <form action="InsertProductServlet" method="post">
            <label for="name">Name:</label>
            <input type="text" id="name" name="name"><br><br>

            <label for="category">Category:</label>
            <input type="text" id="category" name="category"><br><br>

            <label for="price">Price:</label>
            <input type="number" step="any" id="price" name="price"><br><br>

            <label for="quantity">Quantity:</label>
            <input type="number" id="quantity" name="quantity"><br><br>

            <input type="submit" value="Login">
        </form>
    </body>
</html>
```

Source Code Product.java:

```java
package com.lab.model;

public class Product {

    private int id;
    private String name;
    private String category;
    private double price;
    private int quantity;

    // Empty constructor
    public Product() {
    }

    // Parameterized constructor
    public Product(int id, String name, String category, double price, int quantity) {
        this.id = id;
        this.name = name;
        this.category = category;
        this.price = price;
        this.quantity = quantity;
    }

    public Product(String name, String category, double price, int quantity) {
        this.name = name;
        this.category = category;
        this.price = price;
        this.quantity = quantity;
    }

    // Getters & Setters
    public int getId() {
        return id;
    }

    public void setId(int id) {
        this.id = id;
    }

```

```java
    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    public String getCategory() {
        return category;
    }

    public void setCategory(String category) {
        this.category = category;
    }

    public double getPrice() {
        return price;
    }

    public void setPrice(double price) {
        this.price = price;
    }

    public int getQuantity() {
        return quantity;
    }

    public void setQuantity(int quantity) {
        this.quantity = quantity;
    }
}
```

Source Code ProductDAO.java:

```java
package com.lab.dao;

import com.lab.model.Product;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.util.ArrayList;
import java.util.List;

public class ProductDAO {

    private String jdbcURL = "jdbc:mysql://localhost:3307/CSE3023";
    private String jdbcUsername = "root";
    private String jdbcPassword = "";

    // Method untuk mendapatkan sambungan Database
    protected Connection getConnection() {
        Connection connection = null;
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            connection = DriverManager.getConnection(jdbcURL, jdbcUsername, jdbcPassword);
        } catch (SQLException | ClassNotFoundException e) {
            e.printStackTrace();
        }
        return connection;
    }

    // CREATE: Masukkan produk baharu
    public void insertProduct(Product product) {
        String sql = "INSERT INTO products (name, category, price, quantity) VALUES (?, ?, ?, ?)";
        try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
            pstmt.setString(1, product.getName());
            pstmt.setString(2, product.getCategory());
            pstmt.setDouble(3, product.getPrice());
            pstmt.setInt(4, product.getQuantity());
            pstmt.executeUpdate();
        } catch (SQLException e) {
```

```java
            e.printStackTrace();
        }
    }

    // READ: Ambil satu produk berdasarkan ID (Untuk Update)
    public Product selectProduct(int id) {
        Product product = null;
        String sql = "SELECT * FROM products WHERE id = ?";
        try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
            pstmt.setInt(1, id);
            ResultSet rs = pstmt.executeQuery();
            if (rs.next()) {
                String name = rs.getString("name");
                String category = rs.getString("category");
                double price = rs.getDouble("price");
                int quantity = rs.getInt("quantity");
                product = new Product(id, name, category, price, quantity);
            }
        } catch (SQLException e) {
            e.printStackTrace();
        }
        return product;
    }

    // READ: Ambil senarai semua produk (Untuk View)
    public List<Product> selectAllProducts() {
        List<Product> products = new ArrayList<>();
        String sql = "SELECT * FROM products";
        try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
            ResultSet rs = pstmt.executeQuery();
            while (rs.next()) {
                int id = rs.getInt("id");
                String name = rs.getString("name");
                String category = rs.getString("category");
                double price = rs.getDouble("price");
                int quantity = rs.getInt("quantity");
                products.add(new Product(id, name, category, price, quantity));
            }
        } catch (SQLException e) {
```

```java
            e.printStackTrace();
        }
        return products;
    }

    // UPDATE: Kemaskini maklumat produk
    public boolean updateProduct(Product product) {
        boolean rowUpdated = false;
        String sql = "UPDATE products SET name = ?, category = ?, price = ?, quantity = ? WHERE id = ?";
        try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
            pstmt.setString(1, product.getName());
            pstmt.setString(2, product.getCategory());
            pstmt.setDouble(3, product.getPrice());
            pstmt.setInt(4, product.getQuantity());
            pstmt.setInt(5, product.getId());
            rowUpdated = pstmt.executeUpdate() > 0;
        } catch (SQLException e) {
            e.printStackTrace();
        }
        return rowUpdated;
    }

    // DELETE: Padam produk
    public boolean deleteProduct(int id) {
        boolean rowDeleted = false;
        String sql = "DELETE FROM products WHERE id = ?";
        try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
            pstmt.setInt(1, id);
            rowDeleted = pstmt.executeUpdate() > 0;
        } catch (SQLException e) {
            e.printStackTrace();
        }
        return rowDeleted;
    }
}
```

Source Code doPost() InsertProductServlet.java:

```java
protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

    // Get parameters from the form
    String name = request.getParameter("name");
    String category = request.getParameter("category");
    double price = Double.parseDouble(request.getParameter("price"));
    int quantity = Integer.parseInt(request.getParameter("quantity"));

    // Create Product object
    Product newProduct = new Product(name, category, price, quantity);

    // Call DAO to insert product
    ProductDAO productDAO = new ProductDAO();
    productDAO.insertProduct(newProduct); // Panggil DAO, bukan JDBC langsung

    // Redirect to view all products
    response.sendRedirect("ViewProductServlet");
}
```

Source Code doGet() DeleteProductServlet.java:

```java
protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
    int id = Integer.parseInt(request.getParameter("id"));
    ProductDAO.deleteProduct(id); // Panggil DAO
    response.sendRedirect("ViewProductServlet");
}
```

Source Code doGet() ViewProductServlet.java:

```java
protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

    response.setContentType("text/html");
    PrintWriter out = response.getWriter();

    ProductDAO productDAO = new ProductDAO();
    List<Product> listProduct = productDAO.selectAllProducts(); // Panggil DAO

    out.println("<h2>Product List (Using DAO)</h2>");
    out.println("<table border='1'>");
    out.println("<tr><th>ID</th><th>Name</th><th>Category</th><th>Price</th><th>Quantity</th><th>Actions</th></tr>");

    for (Product product : listProduct) {
        out.println("<tr>");
        out.println("<td>" + product.getId() + "</td>");
        out.println("<td>" + product.getName() + "</td>");
        out.println("<td>" + product.getCategory() + "</td>");
        out.println("<td>" + product.getPrice() + "</td>");
        out.println("<td>" + product.getQuantity() + "</td>");
        out.println("<td><a href='UpdateProductServlet?id=" + product.getId() + "'>Edit</a> | ");
        out.println("<a href='DeleteProductServlet?id=" + product.getId() + "'>Delete</a></td>");
        out.println("</tr>");
    }

    out.println("</table>");
    out.println("<br><a href='add_product.html'>Add New Product</a>");
}
```

Source Code doGet() UpdateProductServlet.java:

```java
protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

    response.setContentType("text/html");
    PrintWriter out = response.getWriter();
    int id = Integer.parseInt(request.getParameter("id"));
    Product existingProduct = productDAO.selectProduct(id); // Panggil DAO

    if (existingProduct != null) {
        out.println("<h2>Update Product (Using DAO)</h2>");
        out.println("<form action='UpdateProductServlet' method='POST'>");
        out.println("<input type='hidden' name='id' value='" + existingProduct.getId() + "'>");
        out.println("Name: <input type='text' name='name' value='" + existingProduct.getName() + "' required><br><br>");
        out.println("Category: <input type='text' name='category' value='" + existingProduct.getCategory() + "' required><br><br>");
        out.println("Price: <input type='text' name='price' value='" + existingProduct.getPrice() + "' required><br><br>");
        out.println("Quantity: <input type='text' name='quantity' value='" + existingProduct.getQuantity() + "' required><br><br>");
        out.println("<input type='submit' value='Update Product'>");
        out.println("</form>");
    } else {
        out.println("<h3>Product not found!</h3>");
    }
}
```

Source Code doPost() UpdateProductServlet.java:

```java
protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

    int id = Integer.parseInt(request.getParameter("id"));
    String name = request.getParameter("name");
    String category = request.getParameter("category");
    double price = Double.parseDouble(request.getParameter("price"));
    int quantity = Integer.parseInt(request.getParameter("quantity"));

    Product product = new Product(id, name, category, price, quantity);
    productDAO.updateProduct(product); // Panggil DAO

    response.sendRedirect("ViewProductServlet");
}
```

Output:

Product Form

Name:

Category:

Price:

Quantity:

Login

Product Form

Name: Mineral Water

Category: Beverage

Price: 9

Quantity: 40

Login

Product List (Using DAO)

ID	Name	Category	Price	Quantity	Actions
4	Mineral Water	Beverage	9.0	40	Edit | Delete

Add New Product